Report

Of

Data Science Master Virtual Internship

Submitted

To

School of Computer Science and Engineering

IILM University, Greater Noida, U.P.

In partial fulfilment of the requirement of degree of

B.Tech CSE

By

Roll Number of Student: CS-23411269

Name of Student: PRIYANSHU KUMAR

Section: 4CSE1

Batch: 2023-27

CANDIDATE'S DECLARATION

I, Priyanshu Kumar, Roll No. CS-23411269, do hereby declare that the internship report titled "Data Science Master Virtual Internship" has been completed by me to fulfil the requirement for the award of the degree of Bachelor of Technology in Computer Science and Engineering. All the references have been quoted and I have not taken as such any material from any other source. I affirm that this report is my own work and has not been submitted to any other institute for any degree/diploma requirement, and further I shall be solely responsible for any kind of copyright violation in this regard.

Signature

Student Name: Priyanshu Kumar

Roll No.: CS23411269

Section: 4CSE1

ACKNOWLEDGEMENT

I am using this opportunity to express my gratitude to everyone who supported me throughout the "Data Science Master Virtual Internship" Program. I am thankful for their aspiring guidance, invaluably constructive criticism and friendly advice during this work. I am sincerely grateful to them for sharing their truthful and illuminating views on a number of issues related to this work.

I sincerely thank the All India Council for Technical Education (AICTE), Ministry of Education, and EduSkills for organising this National Internship Portal virtual internship, and Altair for supporting the program and providing valuable learning resources.

I would also like to thank the School of Computer Science and Engineering, IILM University, Greater Noida, for giving me the opportunity to undertake this internship as part of my academic curriculum.

I express my gratitude to my parents for their blessings.

Signature

Student Name: Priyanshu Kumar

Roll No.: CS-23411269

INTERNSHIP COMPLETION CERTIFICATE

शिक्षा मंत्रालय
MINISTRY OF EDUCATION

NATIONAL INTERNSHIP PORTAL

EduSkills
Nation Building Through Skills

AICTE - EduSkills 2026 VIRTUAL INTERNSHIP

Certificate of Virtual Internship

Priyanshu Kumar

IILM University, Greater Noida

has successfully completed the 10-weeks

Data Science Master Virtual Internship

During January - March 2026

May this Internship learning propel you toward a bright and successful career.

Supported by ALTAIR

Ramesha B.S. Head - Academic Initiatives Altair	Dr. Buddha Chandrasekhar Chief Coordinating Officer (CCO) AICTE, Ministry of Education	Shubhajit Jagadev Chief Executive Officer (CEO) EduSkills

Certificate ID : 44f592a22169a4a7c40c

Student ID: NA

INTERNSHIP GRADE O

GRADE- O (Outstanding):90-100 | E (Excellent):80-89 | A (Very Good):70-79 | B (Good): 60-69 | C (Fair): 50-59 | D (Average): 40-49 | P (Pass): 30-39 | F (Fail): Below 30

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CHAPTER 4

PROJECT DESCRIPTION

4.1 Introduction

Data Science and Data Engineering have emerged as some of the most influential fields in the modern technology-driven world, enabling organisations to collect, process, and derive meaningful insights from large volumes of data. In order to gain practical exposure to this field, I enrolled in the “Data Science Master Virtual Internship”, a 10week program organised under the National Internship Portal by the All India Council for Technical Education (AICTE), Ministry of Education, in collaboration with EduSkills and supported by Altair. The internship was conducted virtually during January–March 2026 and provided a structured curriculum covering the fundamentals of data science, data engineering, and analytics.

As part of the internship, I worked on a project titled “End-to-End Retail Data Engineering Project”. The project focused on designing and implementing a complete data engineering pipeline for retail sales data — starting from raw data ingestion and moving through cleaning, transformation, storage in a relational database, and finally analysis and visualization of business insights. This report documents the knowledge gained, tools explored, and the work carried out on this project during the course of the internship, along with the certificate of completion issued upon successful evaluation.

4.2 Organization Profile

The internship was hosted on the National Internship Portal, an initiative of the Ministry of Education and AICTE aimed at bridging the gap between academic learning and industry requirements by providing students with virtual internship opportunities across emerging technologies.

EduSkills, the implementation partner for this program, focuses on “Nation Building Through Skills” by collaborating with leading technology companies to design and deliver industry-relevant virtual internships to engineering students across India. The Data Science Master Virtual Internship was supported by Altair, a global leader in computational intelligence, simulation, and data analytics technologies, which contributed learning content and tools used throughout the internship.

4.3 Problem Statement

Retail businesses generate large volumes of raw sales data on a daily basis, spread across multiple sources such as CSV files, spreadsheets, and point-of-sale systems. This raw data is often unstructured, inconsistent, and unfit for direct analysis. Without a proper data engineering pipeline, businesses struggle to store this data reliably, clean and validate it, and convert it into a structured form from which meaningful sales and customer insights can be derived. The "End-to-End Retail Data Engineering Project" was undertaken to address this problem by building a complete pipeline that ingests raw retail data, processes it through ETL stages, stores it in a structured database, and presents the resulting insights through interactive dashboards.

4.4 Project Objectives

- To design and implement an end-to-end data engineering pipeline for retail sales data.
- To extract raw retail data and load it into a structured PostgreSQL database.
- To clean, transform, and validate the data to ensure accuracy and consistency (ETL).
- To design an appropriate data model/schema for efficient storage and querying of retail data.
- To perform sales and customer analysis using SQL queries and aggregations.
- To visualize key business insights such as sales trends and customer behaviour using Power BI.
- To gain hands-on experience with industry-relevant data engineering tools and practices.

4.5 Scope of the Project

The scope of the "End-to-End Retail Data Engineering Project" covers the complete lifecycle of retail sales data — starting from data extraction from raw CSV/Excel sources, followed by data cleaning, transformation, and validation, loading the processed data into a PostgreSQL database, performing SQL-based sales and customer analysis, and finally presenting the insights through Power BI dashboards. The pipeline and concepts developed in this project are applicable to real-world retail and ecommerce businesses that need reliable, structured, and analysis-ready data for decision-making.

4.6 Technologies and Tools Used

- Programming Language: Python
- Python Libraries: Pandas, NumPy, SQLAlchemy, Psycopg2
- Data Sources & Handling: CSV datasets, Microsoft Excel, Retail Sales Data
- Database: PostgreSQL
- Data Engineering Concepts: ETL (Extract, Transform, Load), Data Cleaning, Data Transformation, Data Validation, Data Modeling
- Data Analysis: SQL Queries, Aggregation, Sales Analysis, Customer Analysis
- Data Visualization: Power BI
- Development Environment: Jupyter Notebook / VS Code
- Version Control: Git and GitHub
- Learning Platform: AICTE EduSkills Virtual Internship Portal (supported by Altair)

4.7 System Architecture

The architecture of the “End-to-End Retail Data Engineering Project” follows a typical ETL-based data pipeline consisting of the following stages: Raw Retail Data (CSV/Excel) → Extraction → Data Cleaning & Transformation → Data Validation → Loading into PostgreSQL Database → SQL-based Sales & Customer Analysis → Power BI Dashboards & Visualization. Python scripts, using libraries such as Pandas, NumPy, SQLAlchemy, and Psycopg2, were used to extract and transform the raw data before loading it into the PostgreSQL database, from where Power BI connected directly to generate interactive reports and dashboards.

4.8 Methodology

The project was carried out in a structured, step-by-step manner following standard data engineering practices. The methodology adopted included:

- Collecting and understanding the raw retail sales dataset (CSV/Excel format).
- Extracting the data using Python and performing initial data profiling.

- Cleaning and transforming the data — handling missing values, duplicates, and inconsistent formats (ETL).
- Designing a suitable database schema/data model for the retail data.
- Loading the transformed data into a PostgreSQL database using SQLAlchemy and Psycopg2.
- Writing SQL queries to perform aggregation, sales analysis, and customer analysis.
- Connecting Power BI to the PostgreSQL database to build interactive dashboards and visual reports.
- Maintaining version control of the project code using Git and GitHub.

4.9 Expected Outcomes

Upon successful completion of the "End-to-End Retail Data Engineering Project" and the internship, I was able to achieve the following outcomes:

- A working end-to-end data engineering pipeline for retail sales data, from raw data to dashboards.
- Practical skills in ETL processes — data cleaning, transformation, and validation.
- Hands-on experience in designing and querying a PostgreSQL database using SQL.
- Ability to perform sales and customer analysis using SQL queries and aggregations.
- Skills in building interactive dashboards and visual reports using Power BI.
- Familiarity with industry-relevant data engineering tools such as Git, GitHub, and Python database libraries.
- Successful completion of the internship with an overall grade of "O – Outstanding".

4.10 Certificate of Completion

The Certificate of Virtual Internship issued by AICTE, EduSkills, and Altair upon successful completion of the "Data Science Master Virtual Internship" has been attached in Chapter 3 of this report as proof of completion. The certificate bears the Certificate ID: 485cb1bb924e7f2d3743 and reflects an overall grade of "O (Outstanding)".

5. BIBLIOGRAPHY / REFERENCES

1. AICTE EduSkills National Internship Portal – Data Science Master Virtual Internship Course Material, 2026.
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3. McKinney, W. "Python for Data Analysis." O'Reilly Media.
4. Documentation of NumPy, Pandas, Matplotlib, Seaborn, and Scikit-learn libraries.
5. Official website of the National Internship Portal, Ministry of Education, Government of India.